IMO 2026 — Autonomous Model Comparison

Independently graded results across seven runs

Compiled July 18, 2026 · "lw" = lightweight single-context harness; "AutoFyn" = multi-agent orchestrator with round-based context resets · grading: six independent verifier agents, IMO 0–7 scale, claims checked symbolically/numerically

Graded scores (not self-reported)

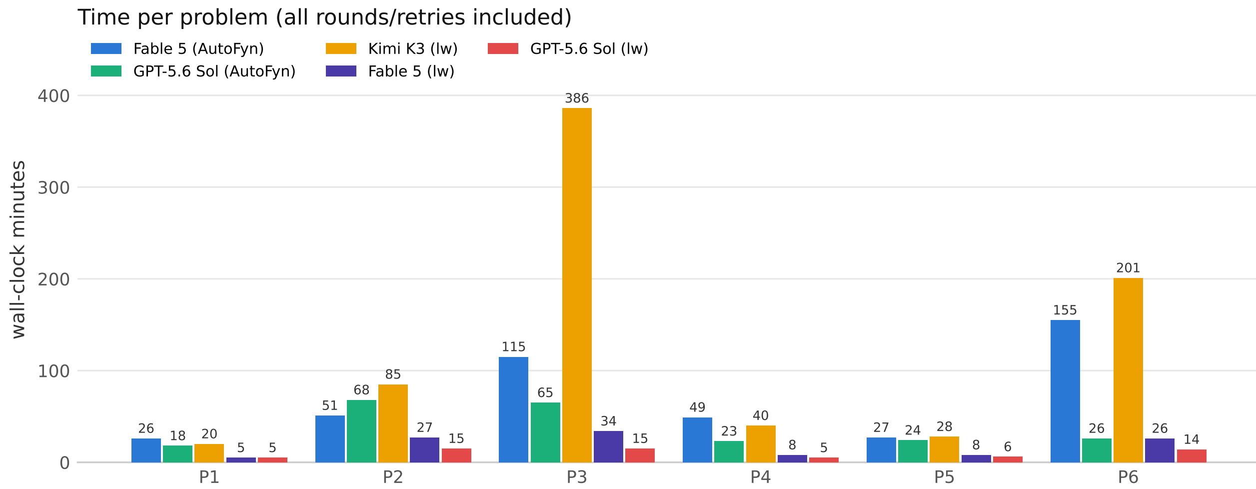
Run	P1	P2	P3	P4	P5	P6	**Total**
Claude Fable 5 (AutoFyn)	7	7	7	7	7	7	**42/42**
GPT-5.6 Sol (AutoFyn)	7	7	7	7	7	7	**42/42**
Claude Fable 5 (lightweight)	7	7	7	7	7	7	**42/42**
Kimi K3 (lightweight)	7	7	5	7	7	3	**36/42**
GPT-5.6 Sol (lightweight)	7	4	2	7	7	1	**28/42**
Claude Opus 4.8 (AutoFyn, partial archive)	7	–	3	–	7	7	24/28
Claude Opus 4.5 (AutoFyn)	6	4	2	5	4	2	**23/42**

Every run except Opus 4.8 (honest "partial" on P3) self-reported all its attempted problems as solved; the grades above reflect independent verification. Notable deductions: Kimi K3's P3 game-value lemma makes a claim refuted by exact minimax and its P6 finiteness crux is proved only for finite transversals; GPT-5.6 Sol (lw)'s P6 central lemma is refuted by an explicit machine-verified counterexample and its P3 rests on an unproven sketch; Opus 4.5 repeatedly asserts cruxes ("SymPy confirms") without proof.

Time and cost

Run	Graded score	Total time	Cost (USD)	Output tokens
Claude Fable 5 (AutoFyn)	42/42	7.1 h	\$32.39	155k
GPT-5.6 Sol (AutoFyn)	42/42	3.7 h	\$5.96	34k
Claude Fable 5 (lightweight)	42/42	1.8 h	\$38.83	371k
Kimi K3 (lightweight)	36/42	12.7 h	∼\$28.20	1,401k
GPT-5.6 Sol (lightweight)	28/42	1.0 h	∼\$4.04	80k

Fable/AutoFyn costs as metered (Anthropic API / OpenRouter). Kimi K3 estimated from logged tokens at list pricing (\$0.30/M cache-hit, \$3/M miss, \$15/M output, 93% hit assumption); GPT-5.6 Sol (lw) from logged tokens at OpenRouter list pricing (\$5/M in, \$30/M out). Kimi K3 time includes all retry rounds (P3 took four).



Findings

- **Three verified 42/42 runs**: both AutoFyn runs and Claude Fable 5 in the bare lightweight harness. Fable 5 lightweight was also the fastest of the three (1.8 h) — it kept full rigor without any orchestration, at 4× the cost of AutoFyn GPT.
- **The harness effect is model-dependent**: GPT-5.6 Sol went from a verified 42/42 inside AutoFyn to 28/42 in the lightweight loop — its 60-minute sweep produced confident write-ups whose crux steps fail under verification on the three hardest problems. Fable 5 showed no such degradation.
- **Kimi K3's 36/42 is real gold-medal-range work** (typical human gold cutoffs are far lower), earned the hard way: 12.7 h including three failed rounds on P3 whose in-context work was lost to poor checkpoint discipline before harness countermeasures landed.
- **Self-reports inflate**: 40 of 41 attempted problem entries claimed "solved"; graders confirmed only 33 as complete (score 7). All graded answers to the four compute-and-prove problems agree across all runs.

Caveats

- Graders are Claude-based agents; despite adversarial instructions, machine-checked counterexamples, and symbolic verification, they are not human medalists. The AutoFyn day-1 solutions (P1–P3) were separately verified by three past IMO medalists.
- Harnesses differ across runs (AutoFyn orchestration vs. single-context loop); times/costs are indicative, not controlled. Prompt-level differences: the lightweight harness is identical across its three runs (same prompt, tools, caps).
- Full per-turn audit trails for every run, plus per-problem grader verdicts with named defects, accompany this document.